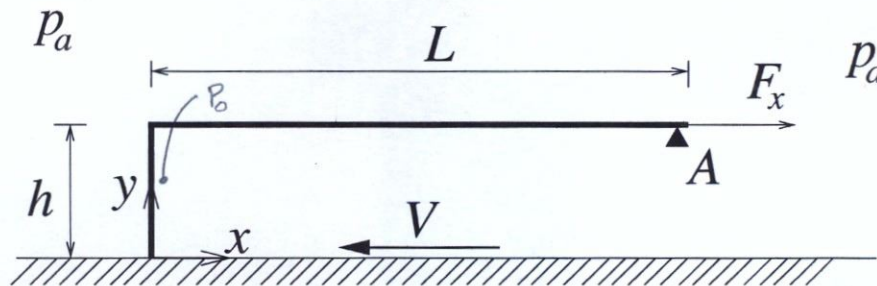


$V=1$

To investigate the performance of windshield wipers consider as a simplified model the L-shaped arm of the figure moving with velocity V with respect to a horizontal surface. The wiper is pulled from the end point A with a force F_x parallel to the wall, dragging in its motion the fluid located between the arm and the wall. To study the problem, we shall use a reference frame moving with the arm, as indicated in the figure. The outer pressure is p_a .

1. Determine the velocity v_x across the channel formed between the wiper and the wall.
2. Find the pressure distribution $p(x)$ including the value found at the end wall $x = 0$.
3. Compute the force F_x necessary to move the wiper.
4. Obtain the moment exerted with respect to the end point A .
5. If W is the weight of the wiper, determine the critical value of V beyond which the end $x = 0$ rises, letting the liquid flow out.



1)

$$v_x = -V \frac{h-y}{h} + \frac{P_0}{2\mu} \frac{h^2 y}{h} \left(1 - \frac{y}{h}\right), \quad Q = \int_0^h v_x dy = \frac{P_0 h^3}{12\mu} - V \frac{h}{2} = 0 \Rightarrow P_0 = \frac{6\mu V}{h^2} = \frac{P_0 - P_a}{L}$$

3)

$$\left(- \int_{\Sigma} (P - P_a) \bar{n} d\sigma + \int_{\Sigma} \bar{E}^1 \cdot \bar{n} d\sigma \right)_x = - \frac{6\mu V L}{h^2} h \bar{e}_x - \mu \left(\frac{dv_x}{dy} \right)_{y=h} L \bar{e}_x = -4\mu V L \bar{e}_x$$

$$F_x = 4\mu V L \frac{L}{h}$$

$P_0 = P_a + \frac{6\mu V L}{h^2}$
 $P - P_a = \frac{6\mu V}{h^2} (L - x)$

4)

$$\bar{M}_{F \rightarrow A} = - \int (\bar{x} - \bar{x}_0) \wedge (P - P_a) \bar{n} d\sigma + \int (\bar{x} - \bar{x}_0) \wedge \bar{E}^1 \cdot \bar{n} d\sigma = - \int_0^L (P - P_a) (L - x) dx \bar{e}_z = -2\mu V L^3 \frac{\bar{e}_z}{h^2}$$

5)

$$\bar{M}_A = \bar{M}_{F \rightarrow A} + W \frac{L}{2} \bar{e}_z - RL \bar{e}_z \rightarrow \text{THE WIPER BEGINS TO RISE WHEN } R=0$$

$V = \frac{W h^2}{4\mu L^2}$